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1st Week Assignment

1. Collaboration with a lab in a country where important information is covered up can lead to adverse consequences. If important information is covered up, it will be very improbable to investigate the origins and causes of a problem as to be able to remediate it and prevent it, especially if there is a serious problem emerging from the lab which can impact lives. For example, it is debated whether the COVID-19 virus emerge from natural causes or from a laboratory leak of a gain-of-function-research on coronaviruses at Wuhan Institute of Laboratory. According to *The Sunday Times*, "Researchers who have attempted to find conclusive proof have been hampered by China's lack of transparency". Restrictions by China for its people to speak the truth include a researcher forbidden to speak out about her research (*The Sunday Times*), a researcher at the Wuhan Institute of Virology suddenly vanishing, and a citizen recording his true experience in Wuhan disappeared (*Sky News Australia, "What Really Happened in Wuhan", YouTube video)*. Also, China intentionally concealed the real site where Wuhan researchers collect virus from bat droppings, although this is an important place to investigate for the origins of COVID-19. This concealment of events made it hard for researchers and intelligence agencies to see and understand the real chain of events leading to the coronavirus pandemic, even though this understanding is crucial since the coronavirus pandemic has impacted the globe and killed millions. The possibility that COVID-19 originated from a lab is a paramount issue as it raises ethical aspects concerning possibility of bioweapons and the dangerous aspects of genetic engineering., and how using these can impact humans worldwide.

   Also, collaboration with a lab in a country with a lack of transparency is a breach of science since science is a quest and explanation for truth. Science is factual and objective. If there is a lack of transparency, this will result in not a science, but rather, a politicization of science: science that does not benefit humanity, but science designed to only benefit certain parties with power and self-interests. The lack of transparency around the COVID-19 research in Wuhan, the fact the research was funded by and related to the military, and the fact that the research was a gain-of-function raised suspicion that the research was created to produce a bioweapon for the government. Not for the people, but for the government. This is the opposite of the aim of science: knowledge that should be open and can be used by the people.

2. Yes, I agree to restrict dangerous research activities by life scientists. Research that will not benefit humanity and instead can harm humans should be banned. One of the types of research that should be given restrictions is gain-of-function research, or "research activities reasonably anticipated to increase a potential pandemic pathogen's transmissibility, pathogenesis, ability to overcome immune response, or ability to overcome a vaccine or drug" (*Written Testimony of Richard H. Ebright).*One of the aspects which raise doubts about the utility of gain-of-function research is that the gain-of-function research is not needed for the protection of human health, but is done simply because it is simpler and faster than creating vaccines. In Ebright's own words in the testimony concerning gain of research, "…gain-of-function research of concern has no civilian practical applications. In particular, gain-of-function research of concern is not needed for, and does not contribute to, the development of vaccines and drugs. (Companies develop vaccines and drugs against pathogens that exist and circulate in humans. Not against pathogens that do not yet exist and do not yet circulate in humans.) Gain-of-function research of concern is performed because it is easy and fast (much faster and much easier than vaccine or drug development) and because, it is fundable and publishable. Not because it is needed." Life scientists need to carefully consider not only to research for the sake of knowledge, but also need to realise the possible impact of the pathogens and/or other organisms they engineer on people and other creatures if the engineered organisms are accidentally released to the outside world.

Another restriction life scientists should face in their research is the safety level of their lab. The Wuhan Institute of Virology lab had been critiqued by US science diplomats from US Embassy in Beijing after they inspect the lab as having " …a serious shortage of appropriately trained technicians and investigators needed to safely operate this high-containment laboratory,". (*Washington Post, Opinion Article, "State Department cables warned of safety issues at the Wuhan lab studying bat coronaviruses, published online April 14, 2020).* If the lab is not sufficiently safe, lab leaks and contamination may occur more easily and can lead to fatalities. The lab should be safe enough by an objective expert control board. To avoid conflict of self-interest, the lab should <u>not</u> be judged safe by the organisation which fund the research.

All life scientists performing dangerous research activities and the funders of the research should account for and be responsible for if there is any lab leak or if they accidentally spread the researched and engineered organism to someone else. Although lab leak may be unlikely, it may happen, as in the case of Wuhan Virology Lab's COVID-19 (still controversial), and the lab leak of lab-created bird virus from a research team in University of Wisconsin and Kawaoka Lab due to human error (factual) (*USA Today, Leak Accident h5n1Virus Avian Flu Experiment*.) These events show that, yes, lab leaks do happen, and we need to raise ethical issues and impose restrictions on dangerous life research. Research

on pathogens should defend and benefit humanity, not to succumb humans to illness and death.

Sources:

1. From reference material in manaba (Sunday Times)
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4. https://www.hsgac.senate.gov/subcommittees/etso/hearings/revisiting-gain-of-function-research-what-the-pandemic-taught-us-and-where-do-we-go-from-here (US Senate Hearing, Aug 2022)
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